

MATTHEW T. LUONGO

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EDUCATION

Scripps Institution of Oceanography University of California, San Diego	<i>La Jolla, CA</i>
Ph.D., Oceanography	Jun. 2024
M.S., Oceanography	Dec. 2020
Harvard University	<i>Cambridge, MA</i>
A.B., Earth & Planetary Sciences, <i>Magna Cum Laude with Highest Honors</i>	May 2017
A.B., Engineering Sciences	

EMPLOYMENT

NOAA Pacific Marine Environmental Laboratory Research Physical Scientist	2026 - Seattle, WA
University of Washington CICOES Postdoctoral Research Fellow	2024 - 2026 Seattle, WA
Scripps Institution of Oceanography, UCSD Graduate Student Researcher	2019 - 2024 San Diego, CA
Wildlands Trust Research Assistant	2018 - 2019 Plymouth, MA
West Monroe Partners Energy & Utilities Experienced Consultant	2017 - 2018 New York, NY

RESEARCH PUBLICATIONS

Published & In Press

15. Xie, S.-P., Miyamoto, A., Peng, Q., & **Luongo, M.T.** (2026). Internal Variability Dominates Tropical Pacific Warming Pattern During Satellite Era. *Proceedings of the National Academy of Sciences*, 123 (36), e2615883123. doi: [10.1073/pnas.2615883123](https://doi.org/10.1073/pnas.2615883123).
14. Cai, C., **Luongo, M.T.**, Deppenmeier, A.-L., Thompson, L., & Kessler, W.S. (2026). Double-ITCZ Bias Reduces Southern Hemisphere Influence on Tropics via Oceanic Pathways. *Geophysical Research Letters*, 53, e2026GL122468. doi: [10.1029/2026GL122468](https://doi.org/10.1029/2026GL122468).
13. **Luongo, M.T.**, Xie, S.-P., Eisenman, I., Sun, S., & Armour, K.C. (2026). Explaining the Equatorial Pacific Thermocline Response to Climate Change with a Model Hierarchy. *Journal of Geophysical Research: Oceans*, 131, e2025JC023559. doi: [10.1029/2025JC023559](https://doi.org/10.1029/2025JC023559).
12. Moon, J.-Y., An, S.-I., **Luongo, M.T.**, Yang, Y.-M., Park, S.-E., Xie, S.-P., & England, M.R. (2026). Asymmetric Cross-Hemispheric Climate Response to Extratropical Forcing Mediated by Tropical Pacific Coupling. *Communications Earth & Environment*, 7 (736). doi: [10.1038/s43247-026-03706-6](https://doi.org/10.1038/s43247-026-03706-6).

11. Hahn, L.C., Lutsko, N.J., Eisenman, I., **Luongo, M.T.**, & Xie, S.-P. (2025). Contribution of AMOC Decline to Uncertainty in Global Warming via Ocean Heat Uptake and Climate Feedbacks. *Journal of Climate*, 38 (21), pp. 6245-6259. doi: [10.1175/JCLI-D-24-0752.1](https://doi.org/10.1175/JCLI-D-24-0752.1).
10. **Luongo, M.T.**, Xie, S.-P., Eisenman, I., Sun, S., & Peng, Q. (2025). How the Subsurface Tropical Pacific Responds to Subtropical Surface Cooling: Implications for Cross-Equatorial Transport. *Journal of Climate*, 38 (14), pp. 3313-3331. doi: [10.1175/JCLI-D-24-0440.1](https://doi.org/10.1175/JCLI-D-24-0440.1).
9. Taylor, B.A., Shi, J.-R., Talley, L.D., Xie, S.-P., **Luongo, M.T.**, & Peng, Q. (2025). Warming Band in Southern Ocean's Indian Sector: The Role of Remote Atlantic Buoyancy Forcing via Poleward-Shifting Circulation Response. *Journal of Climate*, 38 (14), pp. 3219-3228. doi: [10.1175/JCLI-D-24-0425.1](https://doi.org/10.1175/JCLI-D-24-0425.1).
8. Peng, Q., Xie, S.-P., Miyamoto, A., Deser, C., Zhang, P., & **Luongo, M.T.** (2025). Strong 2023-2024 El Niño Generated by Ocean Dynamics. *Nature Geoscience*, 18 (6), pp. 471-478. doi: [10.1038/s41561-025-01700-9](https://doi.org/10.1038/s41561-025-01700-9).
7. Wan, J.S., Chen, C.-C.-J., Tilmes, S., **Luongo, M.T.**, Richter, J.H., & Ricke, K. (2024). Diminished Efficacy of Regional Marine Cloud Brightening in a Warmer World. *Nature Climate Change*, 14 (8), pp. 808-814. doi: [10.1038/s41558-024-02046-7](https://doi.org/10.1038/s41558-024-02046-7)
6. **Luongo, M.T.**, Brizuela, N.G., Eisenman, I., & Xie, S.-P. (2024). Retaining Short-Term Variability Reduces Mean State Biases in Wind Stress Overriding Simulations. *Journal of Advances in Modeling Earth Systems*, 16, e2023MS003665. doi: [10.1029/2023MS003665](https://doi.org/10.1029/2023MS003665).
5. Tseng, H.-Y., Hwang, Y.-T., Xie, S.-P., Kang, S.M., Tseng, Y.-H., **Luongo, M.T.**, & Eisenman, I. (2023). Fast and Slow Responses of the Tropical Pacific to Radiative Forcing in Northern High Latitudes. *Journal of Climate*, 36 (16), pp. 5337-5349. doi: [10.1175/JCLI-D-22-0622.1](https://doi.org/10.1175/JCLI-D-22-0622.1)
4. **Luongo, M.T.**, Xie, S.-P., Eisenman, I., Hwang, Y.-T., & Tseng, H.-Y. (2023). A Pathway for Northern Hemisphere Extratropical Cooling to Elicit a Tropical Response. *Geophysical Research Letters*, 50, e2022GL100719. doi: [10.1029/2022GL100719](https://doi.org/10.1029/2022GL100719)
3. Lutsko, N.J., **Luongo, M.T.**, Wall, C.J., & Myers, T.A. (2022). Correlation Between Cloud Adjustments and Cloud Feedbacks Responsible for Larger Range of Climate Sensitivities in CMIP6. *Journal of Geophysical Research: Atmospheres*, 127, e2022JD037486. doi: [10.1029/2022JD037486](https://doi.org/10.1029/2022JD037486)
2. **Luongo, M.T.**, Xie, S.-P., & Eisenman, I. (2022). Buoyancy Forcing Dominates the Cross-Equatorial Ocean Heat Transport Response to Northern Hemisphere Extratropical Cooling. *Journal of Climate*, 35 (20), pp. 3071-3090. doi: [10.1175/JCLI-D-21-0950.1](https://doi.org/10.1175/JCLI-D-21-0950.1)
1. **Luongo, M.T.**, Kurbatov, A.V., Erhardt, T., Mayewski, P.A., McCormick, M., More, A.F., Spaulding, N.E., Wheatley, S.D., Yates, M.G., & Bohleber, P.D. (2017). Possible Icelandic Tephra Found in European Colle Gnifetti Glacier. *Geochemistry, Geophysics, Geosystems*, 18 (11), pp. 3904-3909. doi: [10.1002/2017GC007022](https://doi.org/10.1002/2017GC007022)

Submitted

3. **Luongo, M.T.**, Blanchard-Wrigglesworth, E., Armour, K.C., Hirasawa, H., Cohen, J.T., & Donohoe, A. Mitigating Summertime Marine Heatwaves via Targeted Marine Cloud Brightening. Manuscript submitted. Pre-print: [10.31223/X5KV4T](https://doi.org/10.31223/X5KV4T)
2. Johnson, G.C., **Luongo, M.T.**, & Lyman, J.M. Spatial Variability of 1993 to 2025 Ocean Temperature Trends Shaped by Poleward Shifts of the Subtropical Gyres. Manuscript submitted to *Geophysical Research Letters*.
1. **Luongo, M.T.**, Armour, K.C., & Johnson, G.C. Reconstructing Satellite-Era Southern Ocean Temperature and Salinity Trends within Observational Uncertainty. Manuscript submitted to *Journal of Climate*. Pre-print: [10.31223/X5DV2H](https://doi.org/10.31223/X5DV2H)

NON-REFEREED PUBLICATIONS

1. **Luongo, M.T.** (2024). Understanding the Climate Response to Hemispheric Differences in Radiative Forcing through Coupled Ocean-Atmosphere Interactions. Doctoral dissertation. University of California, San Diego. [Link](#).
2. **Luongo, M.T.** (2017). Comparison and Calibration of Climate Proxy Data in Medieval Europe. Undergraduate thesis. Harvard University. [Link](#).

HONORS & AWARDS

2024–2026	CICOES Postdoctoral Fellowship (CICOES, UW)
2024–2026	Data Science Postdoctoral Fellowship (eScience Institute, UW)
2024	Early Career Researcher Presentation Award (CFMIP Workshop)
2024	Outstanding Student Presentation Award (AGU 2023 Fall Meeting)
2023	Outstanding Student Presentation Award (CalGFD Meeting)
2022–2024	FINESST Graduate Research Fellowship (NASA)
2022	Outstanding Teaching Assistant (SIO, UCSD)
2021	NSF GRFP Honorable Mention (NSF)
2021	Outstanding Student Mentor (SIO, UCSD)
2019–2020	Regents Fellowship (UCSD)
2017	Hoopes Prize for Outstanding Senior Thesis Work (Harvard)
2015	Undergraduate Summer Research Fellow (Harvard University Center for Environment)
2013	National Merit Program Commended Scholar
2013	John Joseph Moakley Scholar
2012	Eagle Scout

PRESENTATIONS

30. NOAA Pacific Marine Environmental Laboratory, Virtual (May 2026). *Invited*.
29. Stony Brook University, School of Marine & Atmospheric Sciences, Stony Brook, NY (Apr. 2026). *Invited*.
28. University of California, Santa Barbara, Earth Research Institute, Santa Barbara, CA (Apr. 2026). *Invited*.
27. University of Washington, Dept. Atmospheric & Climate Science, Seattle, WA (Mar. 2026).
26. Ocean Sciences Meeting, Glasgow, United Kingdom (Feb. 2026).
25. North Carolina State University, Dept. Marine, Earth & Atmospheric Sciences, Raleigh, NC (Feb. 2026). *Invited*.
24. Columbia University, Dept. Earth & Environmental Sciences, New York, NY (Jan. 2026). *Invited*.
23. AGU Fall Meeting, New Orleans, LA (Dec. 2025).
22. University of Washington, School of Oceanography, Seattle, WA (May 2025).
21. University of Washington, eScience Institute, Seattle, WA (Apr. 2025).
20. Columbia University, Dept. Applied Physics & Applied Mathematics, New York, NY (Feb. 2025). *Invited*.
19. AGU Fall Meeting, Washington, DC (Dec. 2024). Poster.
18. Physical Oceanography Dissertations Symposium (PODS) XIII, Lihue, Kauai, HI (Oct. 2024). *Invited*.

17. National Taiwan University, Dept. Atmospheric Sciences, Taipei, Taiwan (Sep. 2024). *Invited*.
16. University of Tokyo, AORI, Kashiwa, Japan (Aug. 2024). *Invited*.
15. University of Tokyo, RCAST, Tokyo, Japan (Aug. 2024). *Invited*.
14. CFMIP, Boston, MA, USA (Jun. 2024).
13. University of Washington, School of Oceanography, Seattle, WA (May 2024). *Invited*.
12. AGU Fall Meeting, San Francisco, CA (Dec. 2023). Poster.
11. AGU Fall Meeting, San Francisco, CA (Dec. 2023).
10. CalGFD, La Jolla, CA (Sep. 2023).
9. Atlantic Variability and Tropical Basin Interactions Workshop, International Centre for Theoretical Physics, Trieste, Italy (Aug. 2023). Poster.
8. AGU Fall Meeting, Chicago, IL (Dec. 2022).
7. Graduate Climate Conference, Pack Forest, WA (Oct. 2022). Poster.
6. Scripps Student Symposium, La Jolla, CA (Sep. 2022).
5. Equilibrium Climate Sensitivity & Cloud Feedback Symposium, Virtual (Aug. 2022). *Invited*.
4. CalGFD, Pasadena, CA (Aug. 2022).
3. NOAA Climate Sensitivity Task Force Meeting, Virtual (Mar. 2022). *Invited*.
2. AGU Fall Meeting, New Orleans, LA (Dec. 2021).
1. Graduate Climate Conference, Virtual (Oct. 2020). Poster.

TEACHING & MENTORING

Teaching Assistant

- *Dynamics of the Atmosphere and Climate* (SIO 173), UCSD. 2021. Instructors: Shang-Ping Xie & Ian Eisenman.
- *Frontiers and Foundations of Modern Chemistry: A Molecular and Global Perspective* (PS 11), Harvard University. 2015-2017. Instructors: Jim Anderson & Gregg Tucci.

Guest Lecturer

- *Current Research in Climate Science* (OCEAN 475), UW. 2025. Instructor: Steven Riser.
- *Physics of Climate* (ATMS 321), UW. 2025. Instructor: Dargan Frierson.
- *Numerical Modeling of the Climate System* (SIOC 224), UCSD. 2022, 2024. Instructor: Ian Eisenman.
- *Dynamics of the Atmosphere and Climate* (SIO 173), UCSD. 2023. Instructor: Shang-Ping Xie.

Research Mentor

- Graduate: J.-Y. Moon, (Yonsei), C. Cai (UW), M. Nayak (UW)
- Undergraduate: N. Davis (Colorado College), M. Liang (UW)

Peer Mentor

- J. Wan & L. Wicker (SIO).
- SIO-Ask, mentoring program for underrepresented graduate school applicants (2022-2023).

LEADERSHIP AND SERVICE

Field

- AGU Session Convener: “El Niño-Southern Oscillation and Pantropical Climate Interactions: Mechanisms, Predictability, Impacts, and Projections” (2024-present).
- ECS & Cloud Feedback Virtual Symposium Organizer (2024-present).

- CalGFD Organizing Committee (2023).
- Journal Reviews: *Communications Earth & Environment*, *Journal of Climate*, *Climate Dynamics*, *Geophysical Research Letters*, *Earth's Future*, *JGR: Atmospheres*, *Nature Communications*, *Science Advances*, *Weather & Climate Dynamics*
- Proposal Reviews: *NSF Climate & Large-Scale Dynamics*

Department

- SIO Climate Journal Club: Seminar Organizer (2020-2024).
- SIO Peer Mentor Leadership Team (2021-2022).
- SIO Graduate Student Council (2021-2022).
- SIO Faculty Search Committee (2022).

MEDIA

- UW News (Jul. 2026): *Climate Models Get a Vote of Confidence in a New UW Study Mapping Tropical Ocean Temperature Over Time*. [Link](#).
- AGU Eos (Nov. 2025): *Ocean Tunneling May Have Set Off an Ancient Pacific Cooldown*. [Link](#).
- NPR Shortwave Podcast (Jul. 2025): *The Ocean's Currents are Shifting. Why?* [Link](#).
- Scripps News (Jun. 2024): *Artificial Climate Controls Might Become Ineffective—Because of Climate Change*. [Link](#).

FIELD WORK

Oct. 2024: Ocean Observatories Initiative Coastal Endurance Array Fall Cruise

10 days aboard *R/V Sikuliaq* deploying and recovering moorings and gliders off the coast of WA & OR in service to OOI's long-term coastal time series.

Aug. 2023: Advanced Climate Dynamics Course Summer School

1 day aboard *R/V Porsild* taking sediment cores in Disko Bay, Greenland.